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Two decades of cyanobacterial bloom dynamics in the Great Lakes: insights from multi-mission ocean color sensors

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E-mail: sachi.mishra@noaa.gov**Keywords:** CyanoHABs, great lakes, CI_{cyano} , MERIS, MODIS, OLCI, CyanNet, cyanobacteriaSupplementary material for this article is available [online](#)

Abstract

Cyanobacterial harmful algal blooms (cyanoHABs) pose an ongoing threat to water quality, freshwater ecosystems, economies, and public health around the United States Great Lakes region. This study presents a 25 year satellite-based time series (2000–2024) of cyanoHAB observations across the region, generated using data from multiple ocean color sensors, including Envisat-Medium Resolution Imaging Spectroradiometer (MERIS), Sentinel-3A/B Ocean Land Color Imager (OLCI), and MODIS-Terra. To fill the observation gap between MERIS and OLCI sensors, we used CyanNet—a science-informed deep learning framework—to derive the Cyanobacteria Index (CI_{cyano}) from MODIS-Terra, harmonized with CI_{cyano} products from MERIS and OLCI. This approach enabled the creation of the longest continuous and cross-sensor-consistent record of cyanoHAB observations in the Great Lakes to date. We found that differences between sensors were not solely due to CyanNet model limitations but also to variations in temporal coverage and observational timing of the three satellite missions. Analysis of this multi-decadal dataset reveals spatially variable trends in cyanoHAB intensity, frequency, and bloom extent in the basins having the most prolific, historical cyanoHABs: Western Lake Erie, Sandusky Bay (SNB), Saginaw Bay (SGB), Lake Winnebago (LW), and Green Bay (GB). For example, Lake Erie experienced its most severe bloom in 2011, with a cumulative CI_{cyano} of 95.9 and the largest bloom extent on record (6121 km²). The western basin of Lake Erie shows an overall increasing trend in bloom intensity and extent since 2000; however, the blooms exhibited a high degree of interannual variability. In contrast, SNB and SGB exhibit slight decreasing trends in bloom severity, while GB and LW show negligible changes. These long-term observations underscore the value of harmonized satellite data time series and science-informed deep learning models for monitoring inland water quality, and provide a critical foundation for understanding bloom phenology and managing future cyanoHAB risks in the Great Lakes region.

1. Introduction

Cyanobacterial harmful algal blooms (cyanoHABs) have severely impacted the Great Lakes region for over two decades. Blooms in the western basin of Lake Erie have been especially acute and are attributed to bioavailable phosphorus loads from the Maumee River (Kane *et al* 2014, Stumpf *et al* 2016b). CyanoHABs in Western Lake Erie (WLE) have adversely affected recreational activities, charter fishing, tourism, and the drinking water supply of millions of consumers (Steffen *et al* 2017). In addition, frequent occurrences of cyanoHABs have been reported in Saginaw Bay (SGB) in Lake Huron and Lake Winnebago (LW) (figure 1) with summer blooms dominated by cyanobacteria (Wynne *et al*

2022). Green Bay (GB) in Lake Michigan (figure 1) experiences mixed composition algal blooms, often constituting cyanobacteria, diatoms, and green algae during the summer months (Wynne *et al* 2022). Therefore, continuous monitoring of cyanoHABs at the regional scale is needed to understand changing bloom conditions across multiple water bodies. Satellite observations can provide the spatial and temporal resolution needed to monitor these blooms (Stumpf *et al* 2016b). Further, generating long-term time series of bloom observations is equally important for: (1) quantifying temporal changes and patterns to guide water resource management; (2) assessing effectiveness of water quality mitigation measures (Zeng and Binding 2021); and (3) understanding annual bloom formation, phenology, and environmental forcing factors (Mishra *et al* 2023b).

Microcystis-dominated cyanoHABs from 2002 to 2011 have been well characterized with the Cyanobacteria Index (CI_{cyano}) algorithm using satellite observations from the Medium Resolution Imaging Spectroradiometer (MERIS) onboard the Envisat satellite (Wynne *et al* 2008, Stumpf *et al* 2012, Lunetta *et al* 2015) due to the availability of spectral bands required to optically identify cyanobacteria (i.e. 620, 665, 681, and 709 nm band centers). The MERIS mission abruptly ended in April 2012, causing an initial temporal gap in CI_{cyano} -based cyanoHAB satellite monitoring. The Ocean Land Color Imager (OLCI) on the Copernicus Sentinel-3A (S3A) and Sentinel-3B (S3B) satellites were launched in May 2016 and October 2018, respectively, as the continuity mission to MERIS, and include nearly identical bands as MERIS (ESA 2018), enabling the extension of the CI_{cyano} -based cyanoHAB observations from 2016 to the present.

To fill the cyanoHAB observation gap between 2012 (MERIS) and 2016 (OLCI), Wynne *et al* (2013) computed an analogous CI product for the Moderate Resolution Imaging Spectroradiometer sensor on the NASA Terra satellite (hereafter MODIS; launched in 1999) by replacing the MERIS/OLCI 665 nm, 681 nm band, and the 709 nm band with the MODIS 667 nm, 678 nm, and 748 nm near infrared (NIR) band, respectively, for WLE. However, the MODIS CI algorithm can capture non-fluorescing eukaryotic algae. With MERIS and OLCI, the phycocyanin absorption signature around 620 nm is used as an additional flag to distinguish cyanobacteria from eukaryotes (Lunetta *et al* 2015, Stumpf *et al* 2016a). MODIS has some limitations. Due to the lack of a 620 nm band, the MODIS CI product cannot readily provide this discrimination. In addition, the MODIS 748 nm NIR band is substituted for the critical red-edge band centered at 709 nm found on MERIS and OLCI, with a potential loss of sensitivity due to much stronger water absorption. Finally, the MODIS bands were designed for ocean remote sensing, which have a limited radiometric range. Therefore, several key red/NIR bands saturate over bright targets, including turbid water and dense algal blooms (Zeng and Binding 2021, Mishra *et al* 2023a).

By simple band replacement, generating a MODIS CI_{cyano} that is consistent with MERIS and OLCI CI_{cyano} is also challenging due to the differences in the Full Width Half Maximum (FWHM) and their band centers. Wynne *et al* (2021a) addressed this issue by intercalibrating MODIS CI_{cyano} with MERIS and OLCI CI_{cyano} using a simple linear regression approach. Vicariously calibrated coefficients are expected to perform well when applied to water bodies with optical properties like the ones from which calibration data were obtained. For a more generalizable solution, Mishra *et al* (2023a) recently developed CyanNet—a deep learning framework to model a MODIS CI_{cyano} product consistent with MERIS and OLCI in Lake Okeechobee, FL, USA. CyanNet was trained with data from Lake Okeechobee and independently validated without any additional training in Lake Apopka and Lake George, FL, USA, and in WLE. The CyanNet model is geographically transferable without retuning, as demonstrated by Mishra *et al* (2023a), who showed that it had acceptable skill, consistent with MERIS and OLCI, in lakes in other regions. Thus, CyanNet can be extended to the Great Lakes regions to generate a consistent CI_{cyano} product from 2000 to the present using data across three satellite sensors—MERIS, MODIS, and OLCI.

In this study, we generated a continuous CI_{cyano} time series for the Great Lakes region by applying both the algorithmic improvement to the MODIS CI algorithm and the reprocessing improvement of OLCI data with updated on-orbit sensor calibrations. In addition, due to moderate temporal resolution and cloud obscurity, MERIS produced about 2–3 usable images per week in the Great Lakes region, leading to missed observation of bloom events. We thus used CyanNet to gap-fill observations in the MERIS time series with MODIS, and to extend the CI_{cyano} timeseries back to 2000, thereby creating a near-continuous 25 years of satellite remote sensing cyanoHAB time series from 2000 to 2024 in the Great Lakes region.

Our primary goal was to demonstrate the consistency of CyanNet-derived MODIS CI_{cyano} , and to investigate broad temporal patterns and trends in cyanoHABs across the key basins in the Great Lakes region over the last 25 years. Our specific objectives were to:

- Evaluate CyanNet in the Lake Erie subbasins, Sandusky Bay (SNB), SGB, southern GB, and LW;
- Generate a consistent and continuous time series from MERIS, MODIS, and OLCI sensors from 2000 to 2024; and
- Identify long-term trends in cyanoHAB intensity, occurrence, and extent and report similarities and differences in temporal patterns across these regions.

2. Materials and methods

2.1. Study regions

The study regions included the WLE basin, Central Lake Erie basin (CLE), SNB, SGB in Lake Huron, southern GB in Lake Michigan, and LW; figure 1. Lake Erie, primarily its western basin, is well-known and heavily monitored for its annual cyanoHABs (Stumpf *et al* 2016b, Sayers *et al* 2019, Binding *et al* 2021). CyanoHABs occur frequently in the SNB, adversely affecting recreation and public health (Wynne and Stumpf 2015, Wagner *et al* 2024). GB, a significant arm of Lake Michigan, with the Fox River as the main nutrient source (Lin *et al* 2016), has a long history of degraded water quality, including low oxygen levels, which often lead to hypoxic ‘dead zones’; the proliferation of invasive mussels; and elevated nutrient loading, all of which contribute to its susceptibility of dense, summer cyanoHABs (De Stasio *et al* 2014, Wynne *et al* 2022). SGB, a relatively shallow arm of Lake Huron, which receives its nutrients from a single river, the Saginaw River, and is characterized by near annual cyanoHABs (Wynne *et al* 2022). One similarity of Lake Erie, SGB, and the GB sub-basins is that each has a dominant river contributing a large percentage of the total basin-wide nutrient loading—Maumee River to WLE, Fox River to GB, and Saginaw River to SGB (Sayers *et al* 2016, Wynne *et al* 2022). Finally, LW, as the largest lake in Wisconsin, serves as a vital source of drinking water for several major cities in the surrounding area. It is an integral part of the Winnebago Pool, a chain of four shallow lakes that have become highly eutrophic due to extensive phosphorus inputs originating from their predominantly agricultural watersheds, primarily the Fox and Wolf Rivers (Robertson *et al* 2018). As hypertrophic LW directly feeds GB via the Fox River, it has been hypothesized that LW may be playing a role in exporting large quantities of algae and cyanobacteria through the Fox River into GB (Gons *et al* 2008).

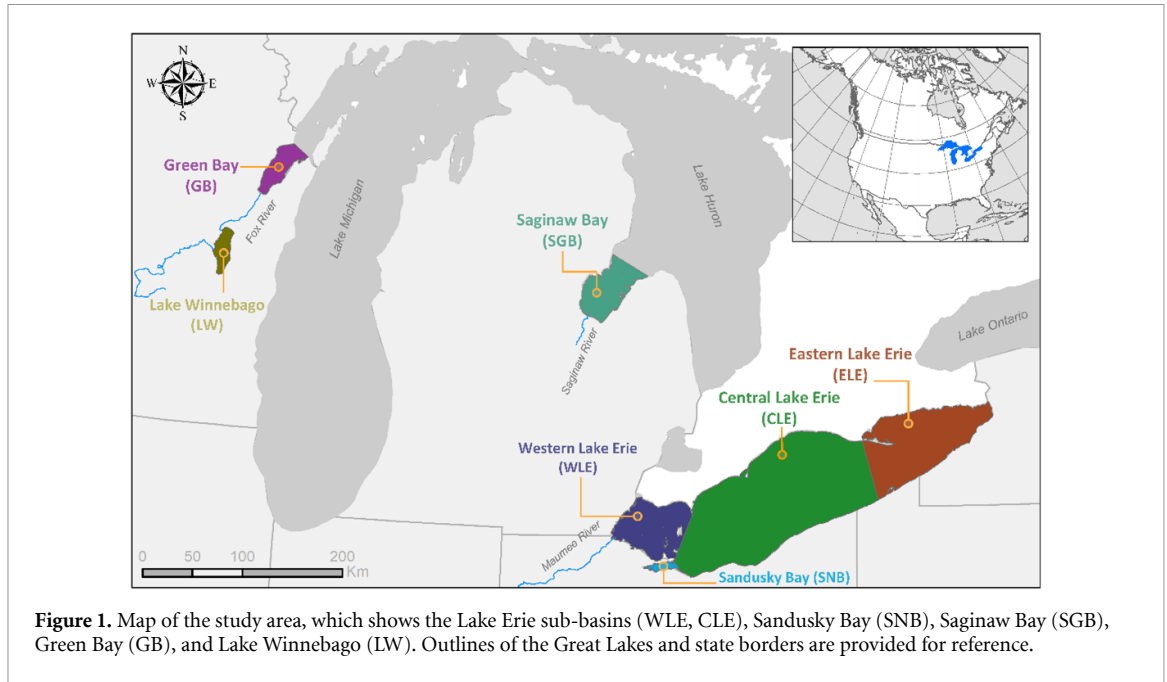
2.2. Satellite data

2.2.1. CI_{cyano} data from MERIS and OLCI

The CI_{cyano} products were generated from data collected by the Envisat MERIS sensor (May 2002 to April 2012) and the Copernicus S3A (April 2016 to present) and S3B OLCI (May 2018 to present) sensors. For MERIS, the reduced resolution (1200 m) data were used from 2002 to 2011. While MERIS has a 300 m full resolution, those data sets were not used as they were not routinely collected over the Great Lakes, and the sparser coverage was redundant with the routine 1200 m coverage. OLCI routinely collects at 300 m, and S3A provided data starting in 2016, while S3B was incorporated starting in 2018. In the *tandem phase* in 2018, S3B had nearly identical orbit times to S3A, allowing cross-calibration. In 2019, the orbits were separated, providing independent and more comprehensive daily data coverage.

We derived spectral (nominal) surface reflectance ($\rho_s(\lambda)$; unitless) from the Level 1B (L1B) data of MERIS and OLCI using NASA’s SeaDAS Ocean Color Science Software (OCSSW) data processing package (SeaDAS 2025). The $\rho_s(\lambda)$ values were obtained by removing Rayleigh radiances corrected for elevation from the instrument-measured top-of-atmosphere radiances. These values were normalized using the downwelling irradiance at the ocean surface to convert them into reflectance. Additionally, cloud and sun glint detection algorithms were used to mask clouds (Wynne *et al* 2018). $\rho_s(\lambda)$ was used instead of the remote sensing reflectance, $R_{rs}(\lambda)$, because the aerosol component of atmospheric correction often fails over turbid productive waters, such as in the Great Lakes regions, reducing valid data retrieval. For example, in WLE, the percentage of valid spectral $\rho_s(\lambda)$ retrievals over water pixels were higher than $R_{rs}(\lambda)$ across all MODIS bands, with a difference of 19% at 412 nm and 15%–20% at other Vis–NIR bands (Zeng and Binding 2021). Additionally, second derivatives implicitly correct for the atmosphere through band differencing (Philpot 1991). As we are using a line height algorithm, which involves band differencing, for bloom detection, these algorithms are also expected to correct for aerosols, making them equivalent to the line height measured from $R_{rs}(\lambda)$ data.

The Cyanobacteria Index (CI), a cyanobacterial biomass indicator was designed around a unique spectral shape centered at 681 nm with reference to a baseline from 665 and 709 nm, referred to as SS(681) (see equation (1)). A negative SS(681) indicates chlorophyll-*a* (chl-*a*) absorption in the absence of chl-*a* fluorescence, which is used as an indicator of the presence of cyanobacteria (Stumpf *et al* 2016a). When chl-*a* fluorescence is present, a cyanobacteria bloom is typically not present, leading to a positive SS(681) and a CI value of 0. As eukaryotic phytoplankton can be present in some of



these water bodies in the summer, and they do not always fluoresce, a second spectral shape, SS(665), (equation (2) was used (Lunetta *et al* 2015). This metric measures the height of the reflectance peak at 665 nm, derived from a baseline connecting the 620 nm and 681 nm bands (equation (1). An elevated presence of phycocyanin-rich cyanobacteria will enhance the reflectance peak at 665 nm, yielding a positive SS(665). Thus, SS(665) serves as a binary flag on top of the CI_{cyano} product ($|SS(681)|$ when $SS(681) < 0$ and SS(665) is positive), confirming the presence of phycocyanin-bearing cyanobacteria (see equation (3)),

$$SS(681) = \rho_s(681) - \rho_s(665) - \{\rho_s(709) - \rho_s(665)\} * \frac{(681 - 665)}{(709 - 665)} \quad (1)$$

$$SS(665) = \rho_s(665) - \rho_s(620) - \{\rho_s(681) - \rho_s(620)\} * \frac{(665 - 620)}{(681 - 620)} \quad (2)$$

$$CI_{cyano} = \begin{cases} |SS(681)| & \text{if } SS(681) < 0 \text{ \& } SS(665) > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

In equations (1) and (2), $\rho_s(\lambda)$ indicates Rayleigh-corrected surface reflectance measured at a band with a band center of λ nm.

S3 instrument calibration performance showed minimal differences in normalized water leaving radiance for S3A and S3B over oligotrophic to mesotrophic waters in the pre-operational *tandem phase*, when both satellites were flying ~ 30 s apart on the same orbit (in 2018) compared to the operational *successive phase* when realistic bio-optical change that can happen in eutrophic waters over the ~ 40 min satellite overpass difference (when not in *tandem*) (Zibordi *et al* 2022). Inter-sensor differences in Zibordi *et al* (2022) due to system vicarious calibration and viewing geometry were relatively small. Compared to field radiometry data, those uncertainties remained within the broader uncertainty envelope over optically complex water for both S3A and S3B. Additionally, because the CI_{cyano} algorithm is a spectral shape algorithm, the uncertainty in the line height measured from the baseline would be minimal. Therefore, we used both OLCI from S3A and B interchangeably to increase the spatial coverage and observation of the blooms.

2.2.2. MODIS data

We obtained Level 0 MODIS Terra imagery from NASA, which was then processed using the SeaDAS OCSSW data processing package to generate Level 3 (L3) Rayleigh-corrected surface reflectance products. We specifically chose to use Terra instead of Aqua because its overpass time is within 90 (and usually less than 60) minutes of MERIS and OLCI. Thus, comparison of Terra with MERIS and OLCI is possible without added temporal differences or uncertainty. Sensors with similar overpass times reduce the uncertainty of cyanoHAB observations due to transport by wind-driven circulation, as well as changes

in the surface concentration caused by the interaction of wind mixing with cell buoyancy especially for *Microcystis* species, which are common in several of the lakes. In addition, Aqua shows performance discrepancies when compared to Terra, with up to a 10% consistent negative bias in global deep water (>1000 m), and temporally anomalous behavior in the critical 667 nm band (NASA 2025). Unlike Sentinel-3A and 3B, which were vicariously intercalibrated during the tandem phase and have closer overpass times, MODIS-Terra and Aqua cannot be used interchangeably without regionally quantifying the uncertainty and performing additional vicarious calibration, which is beyond the scope of this study.

2.3. CyanNet framework

The CyanNet framework includes two components—A logistic regression (LR) classifier and a deep neural network (DNN) quantifier (figure 3 in Mishra *et al* 2023a). Both models are science-informed. The LR model component uses 17 MODIS spectral features (ρ_s and other derived spectral features) to classify a pixel as either cyanobacteria or non-cyanobacteria. Next, the DNN uses 12 spectral features (ρ_s and other derived features) to quantify the magnitude of CI_{cyano} at the cyanobacteria classified pixel (table 2 in Mishra *et al* 2023a). To address the MODIS band saturation issue over bright targets, a replicate of the CyanNet, CyanNet-S, was trained with bands that do not saturate (443, 469, 488, 531, 547, 555, 645, 859, 1240 nm band centers). Thus, CyanNet-S skips the red-NIR bands (667, 678, 748 nm band centers) that tend to saturate over bright targets and returns a CI_{cyano} prediction over the saturated pixels when the standard CyanNet fails. Detailed descriptions of the model components of the CyanNet framework and the lists of spectral features used in CyanNet and CyanNet-S are available in Mishra *et al* (2023a).

In practice, CyanNet generates a CI_{cyano} prediction for each pixel from a MODIS ρ_s L3 file. The file is then processed through CyanNet-S for saturated pixels to obtain CI_{cyano} predictions specifically for any saturated areas. Finally, the predictions from both models are combined to create a unified CI_{cyano} raster (Mishra *et al* 2023a).

2.4. 10 day max compositing

Composite CI_{cyano} images from each of the satellite sensors were created from 10 day periods of daily CI_{cyano} images with no overlap (Wynne and Stumpf 2015), starting on day of year 151 for each year (May 31st in a non-leap year and May 30th in a leap year). We note cyanohABs are typically not observed in these lakes prior to 01 June. Each 10 day composite was made from the maximum (valid) value obtained from each mapped pixel. The compositing method serves two purposes: (1) it helps eliminate most cloud cover over the 10 day periods, and (2) it increases the detection likelihood of cyanohAB biomass. Specifically, *Microcystis* is naturally buoyant and vertically migrates to the surface during calm conditions. Since the satellite primarily measures the surface concentration (within 1 m depth), the compositing method helps capture the areal biomass over a 10 day period more accurately, regardless of wind or other environmental conditions (Stumpf *et al* 2016b).

After generating the 10 day maximum composites from each of the three sensors (MERIS, OLCI, and MODIS) across the study regions, basin-level CI_{cyano} sums for each area, as shown in figure 1, were calculated. The CyanNet CI_{cyano} sums were then compared with the corresponding CI_{cyano} sums from MERIS and OLCI for the lakes or sub-basins.

2.5. Evaluation metrics

10 day max composites from CyanNet were matched with 10 day max composites from MERIS and OLCI for the overlapping periods (2002–2011; 2016–2024). We used median bias and median absolute difference (MedAD) to assess CyanNet performance (Seegers *et al* 2018). Both bias and MedAD were computed from log-transformed data with MERIS or OLCI as the reference or observed and Terra-CyanNet as modeled data.

Median bias quantifies the systematic difference between the modeled value and the observed value, while MedAD better captures the difference in magnitude, highlighting the absolute difference between the observed and modeled CI_{cyano} . A median bias of 1 represents a perfect model with no bias, while a bias of 1.2 indicates a model that overestimates by 20%, and a median bias of 0.8 suggests a model that underestimates by 20%. MedAD numbers are interpreted in the same way as Median bias.

2.6. Bloom extent and frequency

Bloom extent was calculated using a CI threshold of 0.0002 or an equivalent concentration of 20 000 *Microcystis* cells ml^{-1} from the 10 day composites (Lunetta *et al* 2015). In Lake Erie, this value is considered as the concentration when a bloom may pose a potential environmental risk (Davis *et al* 2019).

Table 1. Error metrics from MERIS and MODIS-CyanNet evaluation.

Metrics	Median bias	MedAD	Median bias ^a	MedAD ^a
Western Lake Erie	1.93	2.45	1.36	2.01
Lake Erie Central Basin	0.89	1.99	0.98	2.27
Sandusky Bay	0.83	1.29	0.84	1.27
Saginaw Bay	1.29	1.76	1.34	2.11
Green Bay	1.96	2.04	2.05	2.12
Lake Winnebago	1.06	1.24	1.08	1.22

^a Error metrics from matchup data with 1 km buffer shape.

The number of pixels at or above that threshold was converted to area by multiplying the corresponding pixel areas (km²) of the three sensors.

Bloom frequency or occurrence was calculated using equation (4) in three distinct time periods (2000–2007, 2008–2015, 2016–2024) representing two 8 year periods with the last period containing 9 years of bloom observation,

$$\text{Bloom frequency} = \frac{N_c^{\geq \text{threshold}}}{N_c} \quad (4)$$

where $N_c^{\geq \text{threshold}}$ is the number of composites at or above a CI threshold, and N_c is the total count of 10 day max composites. Bloom frequency varies from 0 to 1, representing no bloom occurrence to 100% bloom occurrence at a given pixel.

2.7. Temporal change analysis

Non-parametric Sen's Slope Estimator and Mann–Kendall's tau were used to assess monotonic trends in time series data. Sen's Slope quantifies the magnitude of the trend (Sen 1968). It is computed from the median of all possible pairwise slopes between data points, providing an unbiased estimate of the slope even in the presence of outliers or non-normality. Together, these methods offer a reliable framework for detecting and estimating trends in time series data.

Kendall's tau statistic indicates the consistency in the trend. It is a ranked correlation coefficient and is determined based on the relative ranking of concordant and discordant data pairs over time (Mann 1945, Kendall 1948). Its value ranges from −1 to 1, with a positive (or negative) value of the test statistic indicating an increasing (or decreasing) trend, respectively. Values greater than |0.3| are indicative of a strong trend. Both Sen's slope and Mann–Kendall's tau were computed using the SciPy package in a Python environment.

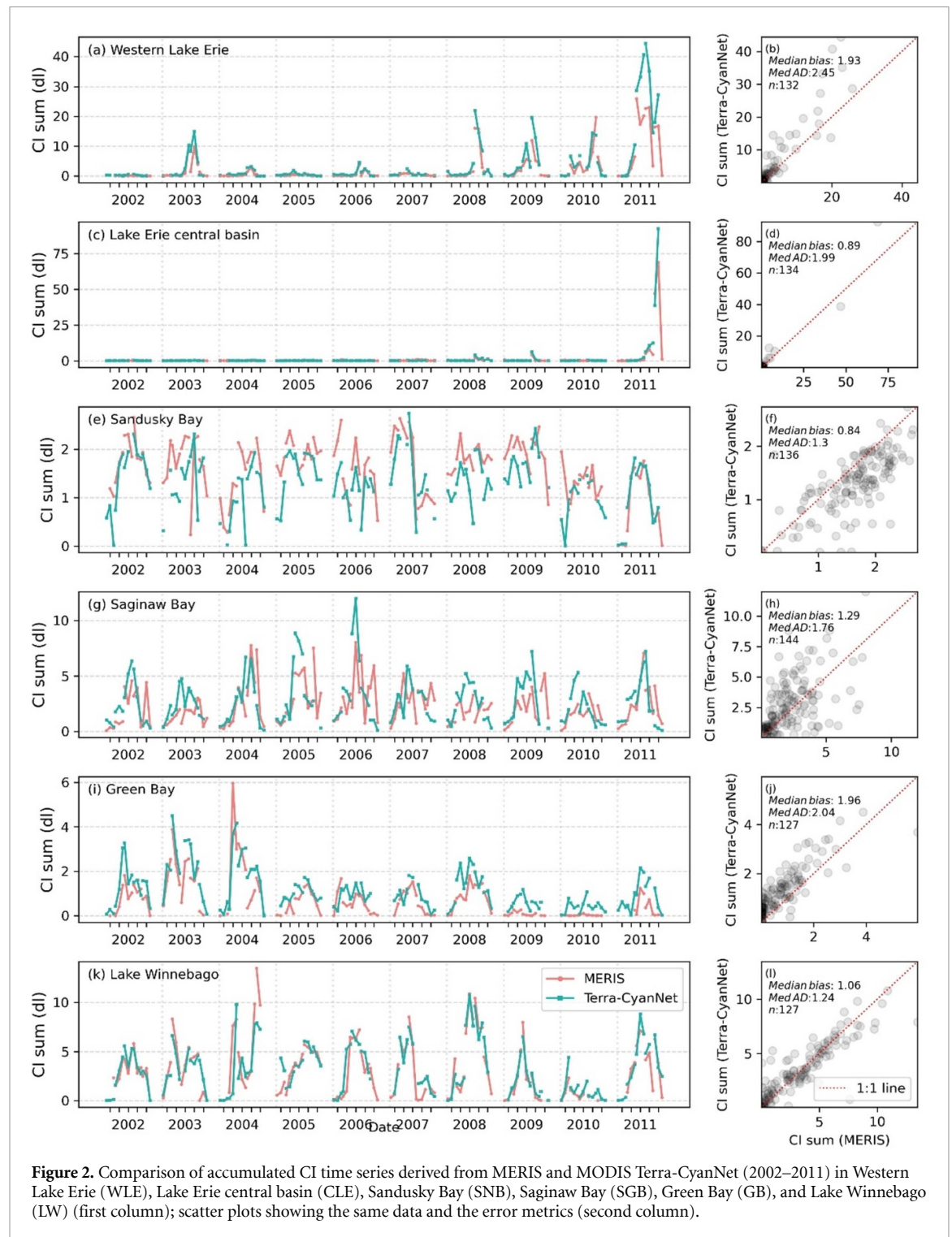
3. Results

3.1. Evaluation of MODIS CI_{cyano} products

A matchup between MERIS CI_{cyano} and CyanNet CI_{cyano} showed that CyanNet was able to retrieve CI_{cyano} consistent with MERIS from 2002 to 2012 in all regions. The median bias varied from 16.5% negative bias in SNB to 96% positive bias in GB (table 1, figure 2). The best matches were observed in LW (MedAD = 1.24) and SNB (MedAD = 1.29). GB and WLE exhibited the greatest differences, with median bias >1.9 and MedAD > 2.0, respectively, indicating that the CI_{cyano} biomass retrievals from CyanNet were larger than those from MERIS in these regions. SGB and CLE fall in the middle, with MedAD between 1.76 and 1.99.

Spatial resolution differences among the three sensors also likely lead to the observed differences, especially when biomass is accumulated along the lake shorelines. Satellites with medium resolution, such as OLCI, can capture the biomass near the shoreline, whereas those pixels would be flagged and removed by MODIS due to the nominal 1.1 km spatial resolution. MERIS integrated 300 m pixels onboard to 1.2 km, so it has potentially better response characteristics than MODIS. Using a 1 km inward buffer (moving the lake boundary 1 km towards the center of the lake) tends to reduce median bias (e.g. WLE: 1.93–1.36) and MedAD (2.45–2.01) but slightly increases MedAD in some lakes (e.g. SGB: 1.76–2.11). Overall, using a 1 km buffered lake polygon appears to improve bias without strongly affecting overall accuracy, especially in spatially heterogeneous systems where the cyanoHAB biomass tends to accumulate near the shoreline.

In the OLCI matchup, WLE, GB, and LW showed substantial reductions compared to MERIS in both bias and MedAD. Like the MERIS-CyanNet matchup, model performance in SNB (MedAD = 45%) and



LW (MedAD = 42%) was stronger than in other regions (figure 3, table 2). Excluding CLE, the maximum difference was observed in SGB (MedAD = 117%) with 27% positive bias. In WLE and GB, the difference appears satisfactory with MedAD values in the range of 1.5–1.56. Use of the 1 km shape reduced MedAD slightly for LW and SNB. However, it increased errors in most other lakes, particularly GB and CLE, implying spatial mismatches due to patchy blooms in the center of these water bodies. The larger differences between OLCI and CyanNet CI_{cyano} were due to differences in the frequency of observation of the missions. Prior to 2019, OLCI had roughly an alternate-day frequency, where MODIS had closer to daily coverage (about 4–5 usable scenes per week, after excluding glint and severe off-nadir views).

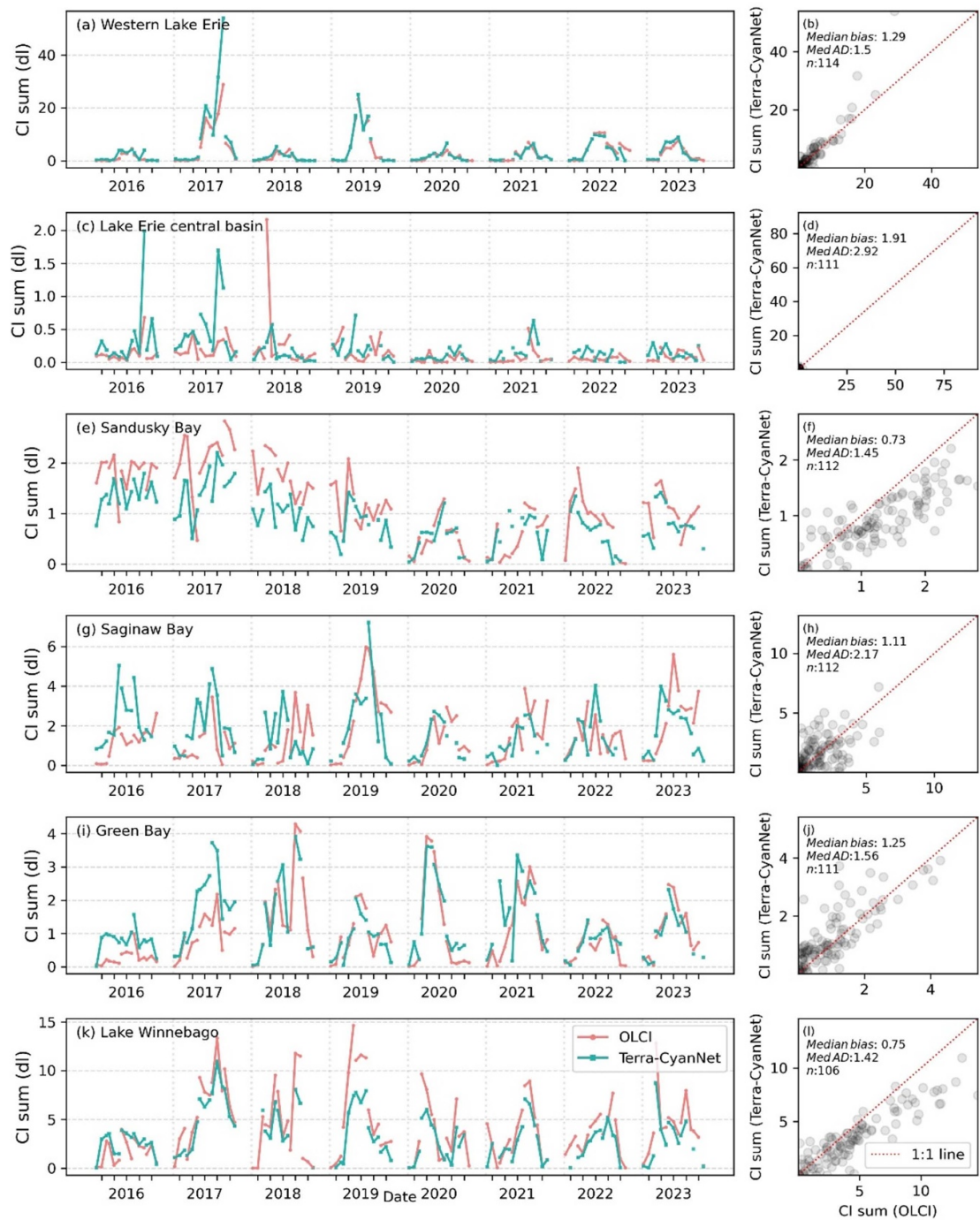


Figure 3. Comparison of accumulated CI_{Cyano} time series derived from OLCI and MODIS-CyanNet (2016–2024) in Western Lake Erie (WLE), Lake Erie Central Basin (CLE), Sandusky Bay (SNB), Saginaw Bay (SGB), Green Bay (GB), and Lake Winnebago (LW) (first column); scatter plots showing the same data and the error metrics (second column).

Table 2. Error metrics from OLCI and MODIS-CyanNet evaluation.

Metrics	Median bias	MedAD	Median bias ^a	MedAD ^a
Western Lake Erie	1.29	1.50	1.26	1.63
Lake Erie Central Basin	1.91	2.93	2.81	3.64
Sandusky Bay	0.73	1.45	0.72	1.43
Saginaw Bay	1.11	2.17	1.19	2.37
Green Bay	1.26	1.56	2.03	2.13
Lake Winnebago	0.75	1.42	0.91	1.29

^a Error metrics from matchup data with 1 km buffer shape.

Starting in 2019, the complementary S3B orbit provided daily coverage by OLCI at 300 m resolution. Hence, after summer 2018 MODIS datasets were no longer included for temporal gap filling due to their lower (1 km) resolution and the additional uncertainty they would introduce.

MedAD in WLE and GB sharply decreased from 95% and 48% respectively, compared to the CyanNet-MERIS matchup, suggesting better consistency between CyanNet and OLCI-derived biomass. However, consistency between the two products somewhat decreased in SNB (15%) and LW (18%).

CyanNet derived CI_{cyano} time series tracked the CI_{cyano} time series from MERIS and OLCI in all regions except SNB, confirming the temporal consistency of MODIS CI (figures 2 and 3). SNB is a smaller water body which may be more prone to complete obscurity from cloud cover. In addition, SNB's *Planktothrix* dominated cyanobacteria community composition (from 2000–2019; Wagner *et al* 2024) may be contributing to the observed difference.

3.2. Continuous CI_{cyano} time series

Evaluation of CyanNet CI_{cyano} sums between MERIS and OLCI demonstrated that CyanNet was able to produce a CI_{cyano} product that was consistent with MERIS from 2002–2011 and OLCI from 2016–2024, providing confidence that CI_{cyano} from MODIS-CyanNet could appropriately fill the observation gap between the MERIS and OLCI missions (2012–2016). In addition, the gaps due to the temporal resolution of the MERIS time series (2–3 d per week) and the OLCI time series prior to the 2019 bloom season could also be filled. Thus, if a 10 day composite was missing in either of the MERIS-OLCI time series, we filled that gap with CyanNet. In addition, if the CyanNet CI sum was significantly larger than MERIS or OLCI pre-2019, we conditionally filled the gap with CyanNet under the assumption that MERIS-OLCI missed the bloom. Our previous validation of CyanNet with daily CI_{cyano} data confirmed that the MedAD was about 29% (Mishra *et al* 2023a) with a 1% positive bias. Thus, a MERIS/OLCI (pre-2019) composite was replaced with CyanNet:

- If MERIS (or OLCI) CI_{cyano} sum is ≥ 50 th percentile of MERIS (or OLCI) CI_{cyano} sum time series, AND
- If CyanNet-MERIS (CyanNet-OLCI) percent difference was $\geq 30\%$.

The first condition guarantees that only the observations during the peak of the bloom season, when biomass was highest, were considered for replacement. This reduces the risk of large fluctuations between CyanNet and MERIS/OLCI during periods of low biomass. The second condition ensured that only significantly different CyanNet retrievals outside of the model error range were considered for replacement, thereby increasing the confidence of MERIS/OLCI replacement by MODIS-CyanNet.

Thus, we filled the Envisat (MERIS) and S3 (OLCI) mission gap from 2012 to 2015 and incorporated the MODIS-CyanNet CI_{cyano} timeseries into the MERIS (2002–2011) and OLCI (2016–2018), as described above. Additionally, the time series was extended back to 2000 by appending MODIS-CyanNet CI_{cyano} from 2000 and 2001, thereby creating a continuous and consistent, 25 year CI_{cyano} time series from 2000–2024 for the Great Lakes region (figure 4).

3.3. Cyanobacteria bloom intensity

Lake Erie: WLE exhibited substantial interannual variability in cyanoHAB intensity, with a pronounced peak in 2011 (mean = 16.8; max = 44.4; figure 5(a)). The most intense bloom in WLE occurred in 2017 (mean = 10.07; max = 53.9). These episodic high-intensity bloom events occurred between 2008 and 2019, in contrast to the generally low values observed during the early 2000s (except 2003). An overall increasing trend in bloom intensity was observed in WLE (Sen's slope = 0.098 CI yr^{-1} ; Kendall's tau = 0.33; table 3). The Sen's slope is expected to lie within 0.051 and 0.271 CI yr^{-1} at a 95% confidence level. As expected, CLE consistently displayed the lowest CI_{cyano} intensity, with a minimum value of 0.05 and a median of annual means of 0.24 (CI unit; figure 5(b)). An extreme bloom intensity was recorded in 2011 (mean = 9.8; max = 68.7) and 2015 (mean = 4.50; max = 22.2), as blooms were advected from WLE into the CLE. Collectively in WLE and CLE, representing the total cyanoHAB biomass in Lake Erie, 2011 was the most severe bloom year with CI of 95.9, followed by 2017 (CI of 55.0) and 2015 (CI of 48.4; figure S1).

SNB: SNB maintained moderate mean bloom intensity throughout the 25 year study period, with limited variation in maximum intensity (figure 5(c)). The results show an overall decreasing trend (Sen's slope = $-0.032 \text{ CI yr}^{-1}$; Kendall's tau = -0.31 ; table 3), with slight increase during the early 2000s

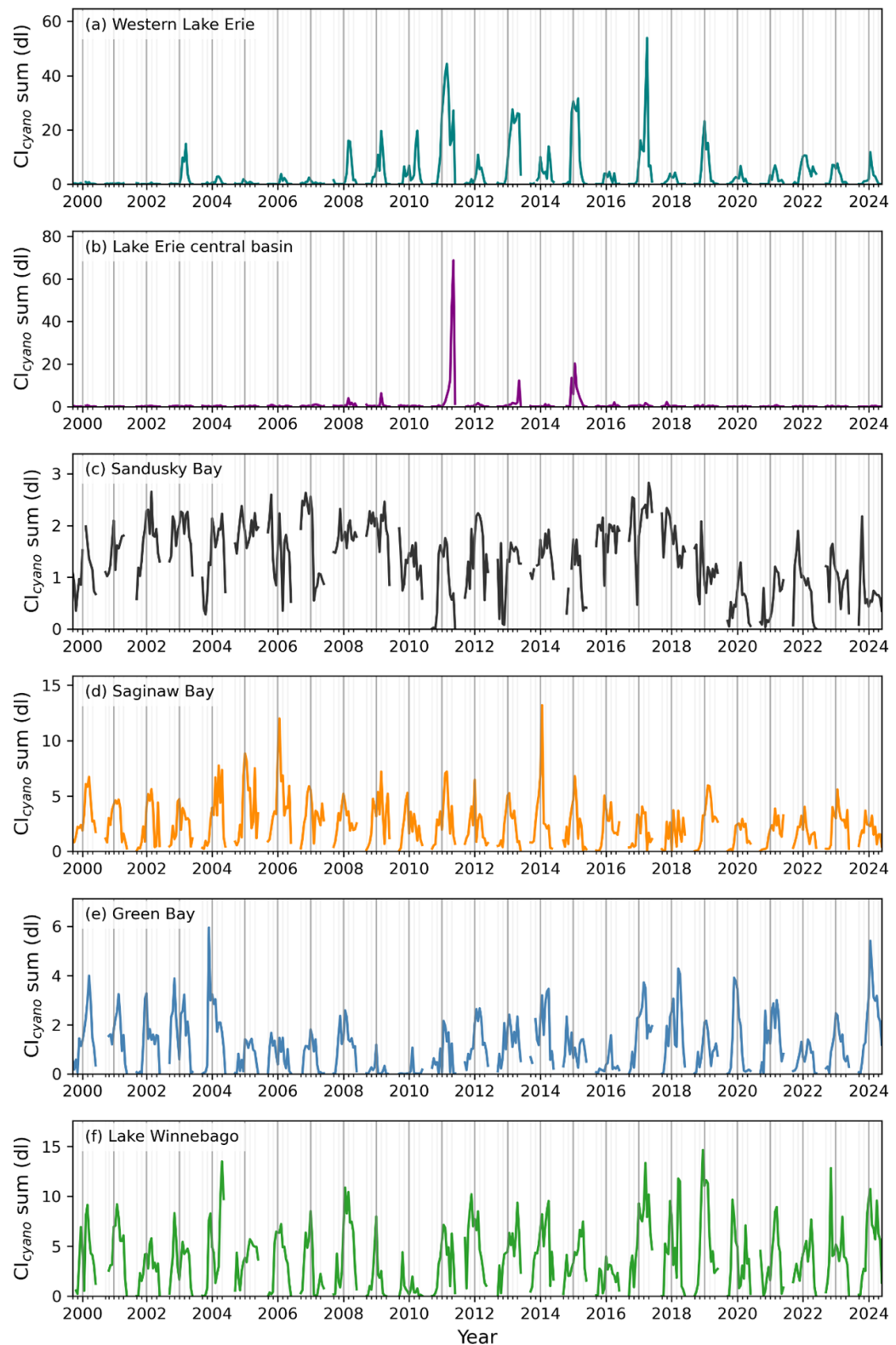


Figure 4. Continuous Cl_{cyano} time series from 2000 to 2024 in the Great Lakes region—(a) Western Lake Erie (WLE), (b) Lake Erie central basin (CLE), (c) Sandusky Bay (SNB), (d) Saginaw Bay (SGB), (e) Green Bay (GB), and (f) Lake Winnebago (LW). Note: Only time series data from June through October for every year are shown here. The vertical gray lines show the position of the midpoint of the bloom season (13 or 14 August, based on leap/non-leap year).

and again in 2017 (mean = 2.09), followed by a decrease most likely due to a dam removal upstream of SNB in 2018 (Wagner *et al* 2024). Overall, the Sen's slope is expected to vary within -0.055 to -0.003 $CI\ yr^{-1}$ at a 95% confidence level.



SGB: SGB showed an overall decreasing trend (Sen's slope = -0.056 CI yr^{-1} with a range of -0.097 to -0.024 CI yr^{-1} ; Kendall's tau = -0.45 ; table 3) with periodic peaks (e.g. max = 12.0 in 2006 and 13.2 in 2014).

GB: GB showed a negligible change in the mean bloom intensity (Kendall's tau = 0.03) over the entire time period. However, after a large decline in 2005, the blooms tended to recover to pre-2005 values by 2024.

LW: LW consistently recorded the highest maximum values across the study regions, with repeated peaks exceeding 10.0 (e.g. 13.48 in 2004; 14.6 in 2019; 12.8 in 2023; figure 5(f)). The lake also sustained elevated mean values, particularly in 2017 (mean = 6.2), indicating persistent high-intensity cyanoHABs.

Table 3. Mann–Kendall test results showing trends in the time series of annual mean bloom intensity and spatial extent in the study regions. Upper and lower bounds of Sen's slope (CI yr^{-1}) represent the 95% confidence interval. Kendall's tau $>|0.3|$ indicates a substantively consistent trend.

Bloom intensity trends				
	Sen's Slope (CI yr^{-1})	Upper/Lower bounds of Sen's slope (CI yr^{-1})	Trend	Kendall's Tau
Western Lake Erie	0.098	[0.051, 0.271]	Increasing	0.33
Lake Erie central basin	−0.002	[−0.015, 0.01]	No trend	−0.04
Sandusky Bay	−0.032	[−0.055, −0.003]	Decreasing	−0.31
Saginaw Bay	−0.056	[−0.097, −0.024]	Decreasing	−0.45
Green Bay	0.003	[−0.03, 0.036]	No trend	0.03
Lake Winnebago	0.029	[−0.033, 0.106]	No trend	0.12
Bloom extent trends				
	Sen's Slope ($\text{km}^2 \text{yr}^{-1}$)	Upper/Lower bounds of Sen's slope ($\text{km}^2 \text{yr}^{-1}$)	Trend	Kendall's Tau
Western Lake Erie	18.75	[10.59, 31.7]	Increasing	0.41
Lake Erie central basin	2.44	[−2.93, 9.81]	No trend	0.12
Sandusky Bay	−0.12	[−1.05, 1.08]	No trend	−0.03
Saginaw Bay	−5.42	[−14.02, 3.89]	No trend	−0.17
Green Bay	−2.32	[−9.44, 6.04]	No trend	−0.09
Lake Winnebago	2.33	[−0.43, 4.93]	No trend	0.26

Overall, WLE and LW were characterized by episodic high-intensity blooms. In contrast, SNB and SGB showed a slight decrease in bloom intensity over the study period. GB exhibited larger blooms at the beginning and end of the time period.

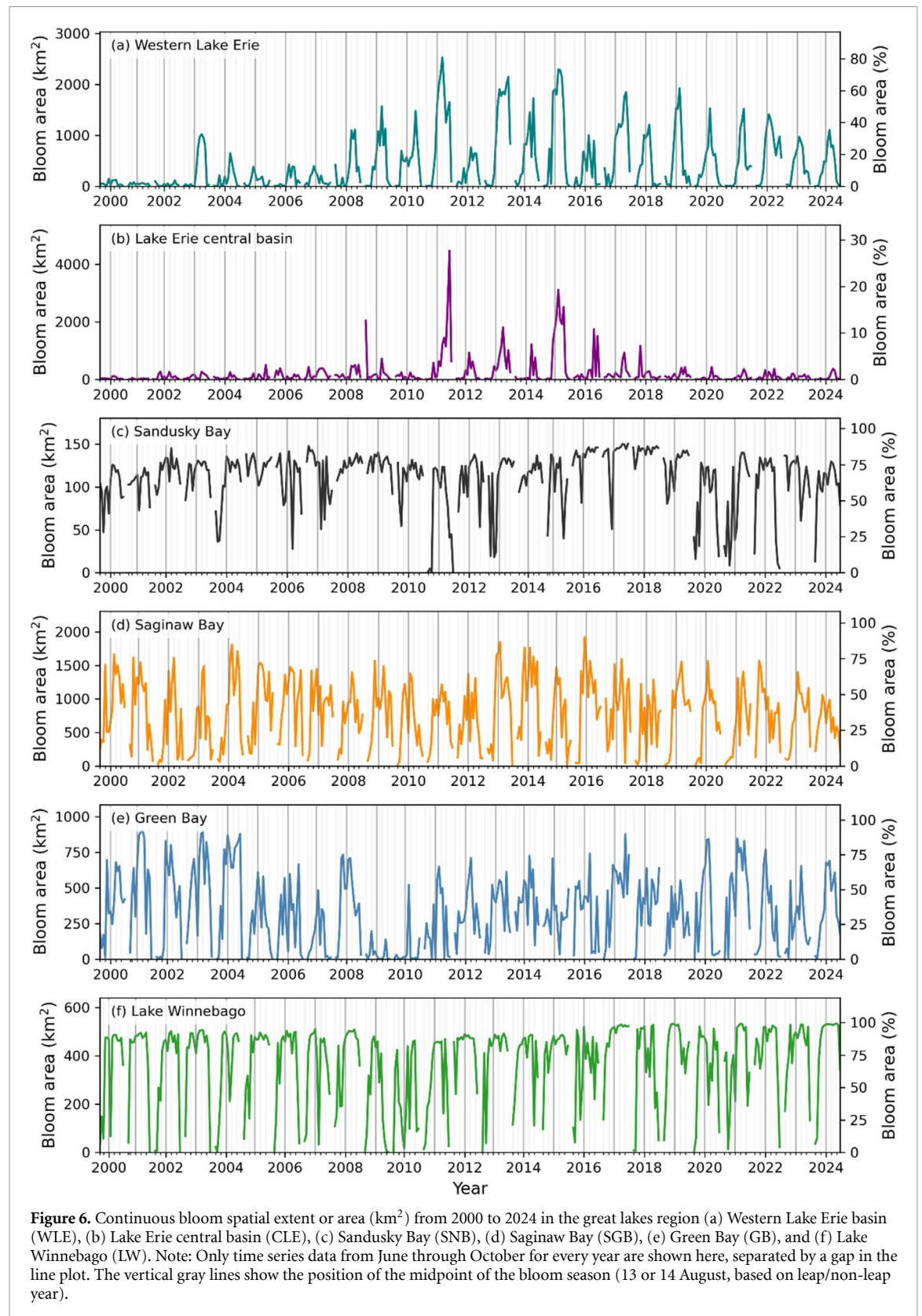
3.4. Cyanobacteria bloom extent

Lake Erie: The extent of cyanobacteria blooms in the WLE was relatively low during the 2000–2002 bloom seasons (e.g. maximum extent of 115 km^2 (3.7% of surface area) in 2002; figure 5(g), figure 6(a), table S1). However, there were sharp increases in bloom extent in 2003 (1020 km^2 ; 32.6% of the surface area), later peaking between 2009–2011 and again from 2013–2015. The mean bloom extent in the WLE substantially increased over the study periods ($18.8 \text{ km}^2 \text{yr}^{-1}$; table 3), reaching its peak in 2011 at 1044 km^2 (33% of the surface area) and in 2015 at 1002 km^2 (32% of the surface area), before slightly declining through 2024. As expected, the CLE does not experience widespread cyanohABs during most years. However, 2009, 2011, and 2015 were notable bloom years, with maximum extents varying from 2041 to 4468 km^2 (12.7–27.7% of the surface area) in the CLE (figure 5(h), table S1). Review of the 10 day time series composites indicates that the larger blooms in these three years did not originate in the CLE; instead, they were likely advected from the WLE. Collectively in both WLE and CLE, 2011 had the largest bloom area (max = 6121 km^2), followed by 2015 (max = 5408 km^2), and 2013 (max = 3667 km^2 ; figure S1).

SNB: In SNB, the bloom extent remained relatively steady, with very low variability, indicating consistent bloom conditions since 2000 (figure 6(c)). For instance, the mean extent increased from 99 km^2 (59.3% of the surface area) in 2000–102 km^2 (60.7% of the surface area) in 2024, with little variation (figures 5(c) and 6(c), table S1). This stability is expected, as cyanohABs have consistently occurred in the entire bay since 2000, reaching the upper limit of the spatial extent. However, the seasonal median extent briefly decreased after 2018, which could be attributed to the decrease in *Planktothrix* population in the bay due to the Ballville dam removal (Wagner *et al* 2024). Since 2021, bloom extent has been returning to the pre-2018 baseline level.

SGB: Bloom extent in SGB demonstrated large inter-annual variability (figure 6(d)). The peak bloom years were notably in 2004 (1813 km^2 ; 84.8% of the surface area), 2013 (1850 km^2 ; 86.5% of the surface area), and 2016 (1922 km^2 ; 89.9% of the surface area), with mean bloom extents ranging from 529 to 782 km^2 (25%–36% of the surface area) (figure 5(j), table S2).

GB and LW: In GB, the maximum bloom extent showed a decreasing trend from 2001 (904 km^2 ; 92.5% of the surface area) to 2009 (197 km^2 ; 20% of the surface area), followed by a gradual increase until



2017 (figure 5(k), table S3). A similar pattern was observed in the mean bloom extent. LW has experienced a subtle increase in bloom extent since 2000 (no meaningful trend; table 3). The mean bloom extent increased from approximately 352 km^2 in 2000 to around 438 km^2 in 2024 (figure 5(f), table S3). Notably, 2009–2011 were characterized by relatively small blooms, with the mean extent decreasing to about $230\text{--}292 \text{ km}^2$ (43%–55% of the surface area).

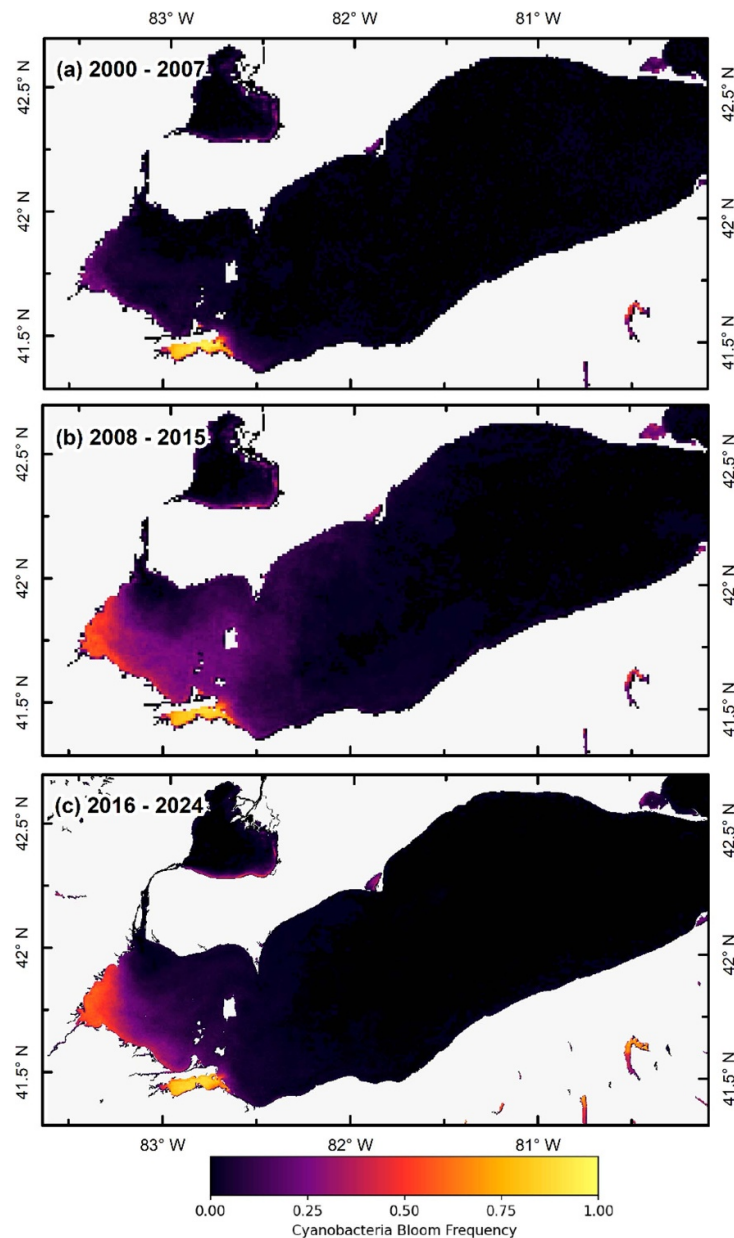


Figure 7. Cyanobacteria bloom frequency in the Lake Erie basins and Sandusky Bay (SNB) during (a) 2002–2011 (MERIS), (b) 2012–2015 (MODIS-CyanNet), and (c) 2016–2024 (OLCI). Pixel-level frequency indicates the occurrence of cyanobacteria with $>20\,000$ cells ml^{-1} in all 10 day max Cl_{cyano} composites within the corresponding observation periods.

3.5. CyanoHAB frequency

In the WLE, bloom frequency increased from 2000–2007 (mean = 0.05) to a peak in 2008–2015 (mean = 0.22), followed by a moderate decline in 2016–2024 (mean = 0.15), suggesting episodic worsening of blooms followed by slight improvement, though bloom occurrences remain elevated compared to the early 2000s (figures 7 and 8). In CLE, bloom frequency increased moderately in 2008–2015 (mean = 0.04) compared to 2000–2007 (mean = 0.02), returning to baseline levels in 2016–2024. CLE shows relatively low and stable bloom frequencies throughout the period. Occasional fluctuations were primarily due to the movement of blooms from the WLE. On the other hand, SNB consistently witnessed high bloom frequencies across all periods, with mean values ranging from 0.65 to 0.68 and median values above 0.75, suggesting persistently severe bloom conditions with little temporal variation.

In Lake Michigan, GB experienced a small decrease in mean bloom frequency from 0.34 (2000–2007) to 0.27 (2016–2024), despite year-to-year fluctuations (figures 8 and 9(a)–(c); year-to-year changes are not shown). Bloom frequency increased slightly over time in LW, with mean values increasing from 0.67 (2000–2007) to 0.73 (2016–2024).

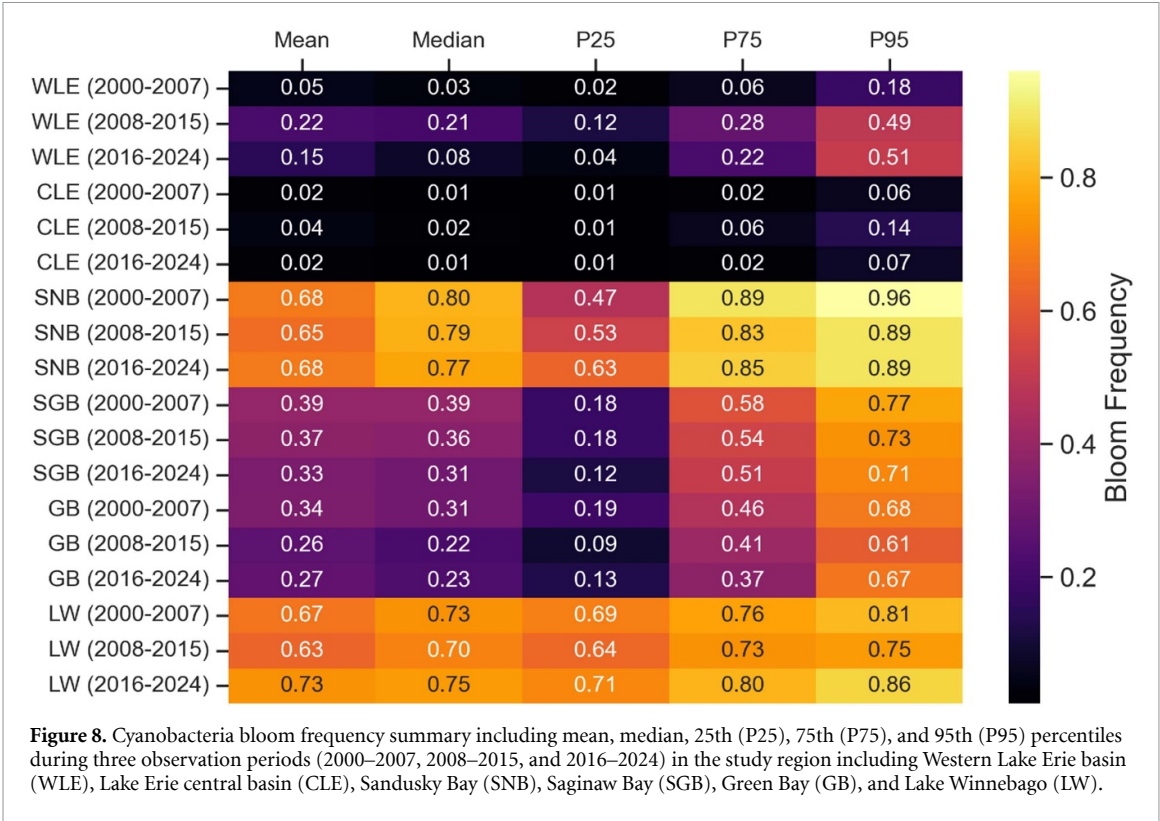


Figure 8. Cyanobacteria bloom frequency summary including mean, median, 25th (P25), 75th (P75), and 95th (P95) percentiles during three observation periods (2000–2007, 2008–2015, and 2016–2024) in the study region including Western Lake Erie basin (WLE), Lake Erie central basin (CLE), Sandusky Bay (SNB), Saginaw Bay (SGB), Green Bay (GB), and Lake Winnebago (LW).

4. Discussion

The frequent occurrence and intensity of cyanoHABs pose a significant and persistent threat to the ecological health, recreational utility, and drinking water safety of large freshwater systems across the globe, including in the Great Lakes region (Stumpf *et al* 2016b). In this study, we developed a multi-sensor, long-term (2000–2024) cyanoHAB time series for the Great Lakes by leveraging data from three satellite missions—Envisat-MERIS, MODIS, and S3A/B OLCI. We achieved this by incorporating and evaluating an existing deep learning Model, CyanNet for MODIS, thereby allowing us to fill gaps in the MERIS/OLCI time series (Mishra *et al* 2023a). The time series provides critical insights into the spatiotemporal dynamics of these blooms in the Great Lakes region, offering a foundational understanding necessary for effective cyanoHAB management and mitigation strategies.

In previous studies, researchers have used multi-sensor data fusion approaches to generate long-term bloom time series (Ho *et al* 2017, Zeng and Binding 2021). For example, MODIS data was used to generate a CI product by replacing MERIS bands with the nearest MODIS bands to rapidly fill the mission gaps between MERIS and OLCI when MERIS was abruptly decommissioned (Wynne *et al* 2013). This adjustment produced MODIS CI data that showed good agreement with MERIS data. However, it was constrained by the saturation of critical bands in turbid water. More recently, Zeng and Binding (2021) successfully used MERIS, MODIS, and OLCI to generate a maximum chlorophyll index time series in Lake Erie, Lake Winnipeg, and Lake of the Woods using a Neural Network approach. Their approach required retraining the neural net models for each water body separately (Zeng and Binding 2021). Sayers *et al* (2019) used NASA's Sea-Viewing Wide Field-of-View Sensor (SeaWiFS) and MODIS to construct 20 years of bloom extent time series in the WLE from 1998 to 2017, while Ho *et al* (2017) used MODIS CI (2012–2015;(Wynne *et al* 2013) to extend the Landsat 5 (1984–2001), and MERIS CI (2002–2011) to produce a long-term phytoplankton bloom extent time series (1984–2015). However, replacing the 709 nm band with the 754 nm band in MODIS CI requires it to be intercalibrated with MERIS and OLCI. Additionally, saturation of red-NIR channels in MODIS over high-scattering waters was a noted issue that needed an improved solution (Wynne *et al* 2013). CyanNet, which leverages the better generalizability characteristics of deep learning algorithms, eliminates the problem of band saturation and does not involve retraining the model for individual water bodies. In addition, it produced a CI_{cyano} product

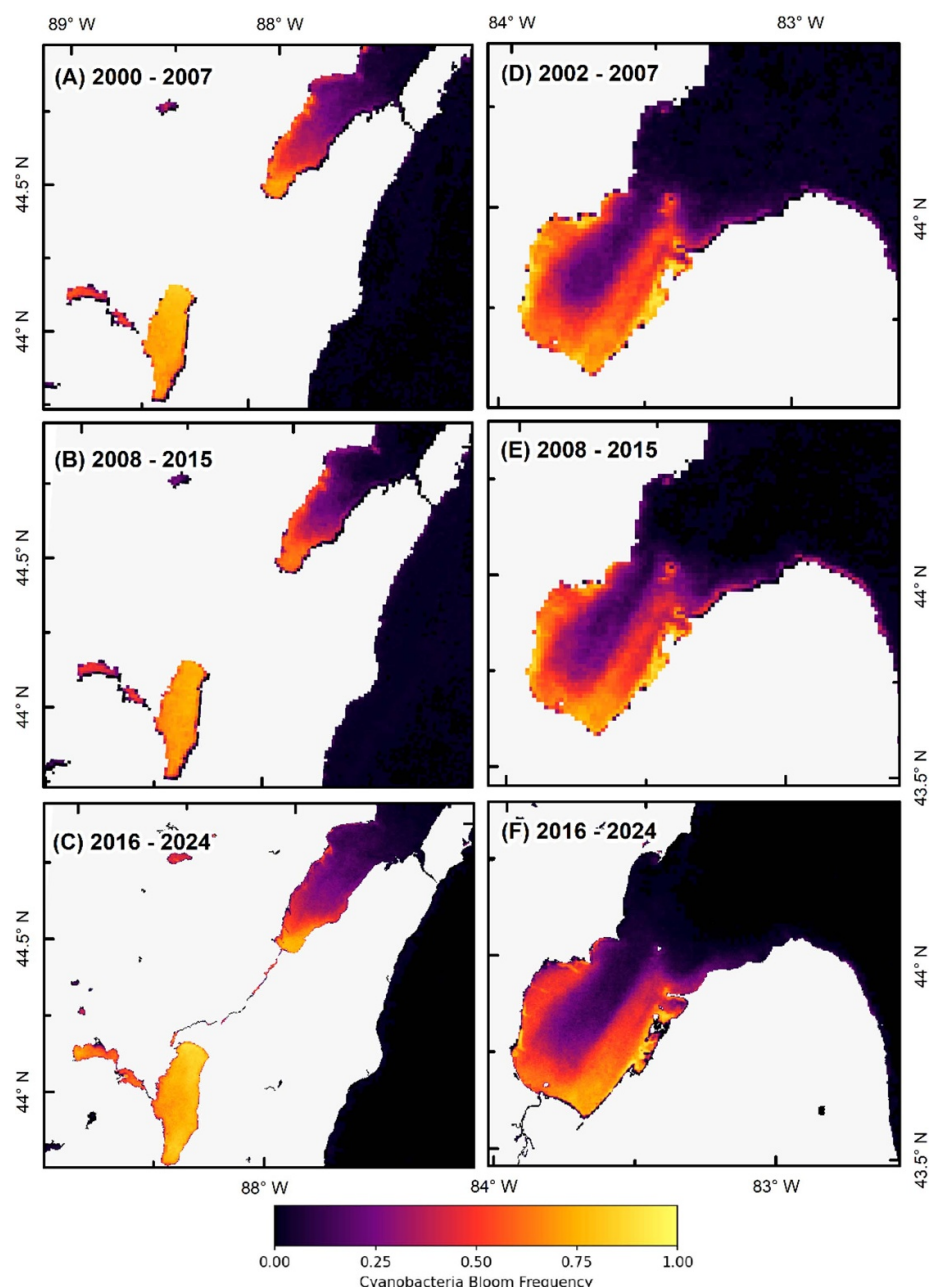


Figure 9. Cyanobacteria bloom frequency in Green Bay (GB) and Lake Winnebago (LW) during (a) 2000–2007 (MERIS), (b) 2008–2015 (MODIS-CyanNet), and (c) 2016–2024 (OLCI). Pixel-level frequency indicates the occurrence of cyanobacteria with $>20\,000\text{ cell ml}^{-1}$ in all 10 day max CI_{cyano} composites within the corresponding observation periods. (d)–(f) Cyanobacteria bloom frequency in Saginaw Bay (SGB) during those three periods.

consistent with MERIS and OLCI, while addressing the red-NIR saturation problem (Mishra *et al* 2023a). Compared to previous efforts, CyanNet produced a more harmonized and consistent CI_{cyano} product that we used to generate a continuous cyanoHAB time series in the Great Lakes region.

4.1. Leveraging CyanNet to increase temporal and spatial bloom coverage

The standard DNN that makes up CyanNet had a MedAD of 27% as observed in an independent test dataset in Lake Okeechobee, FL, from 2009 and 2016 (Mishra *et al* 2023a). The matchup was carried out with daily CI_{cyano} data at the pixel level. Thus, the MedAD reflected the inherent model error and exhibited minimal uncertainties from other extraneous sources such as spatio-temporal differences in satellite coverages, differences in satellite viewing angles, as well as data contamination or spatial data loss due to either cloud cover, sun glint, or adjacency issues (Mishra *et al* 2023a).

In a previous study, CyanNet was applied to WLE to assess its performance in this region without further local training and validation (Mishra *et al* 2023a). Briefly, this evaluation used 26 daily CI_{cyano} image pairs obtained from MERIS/OLCI and Terra-CyanNet. It underwent visual quality control to

exclude clouds, haze, and sun glint, representing the best-case matchup data available. These initial match-ups showed the mean Chl-a concentrations from Terra-CyanNet differed from those of MERIS/OLCI by 29%, with a 9% positive bias. This discrepancy includes model error, differences in sensor characteristics, view angle (atmospheric interference), and bio-optical variations between the overpass times. The MedAD from the daily matchups was similar to that observed in Lake Okeechobee, at 36%. In smaller lakes, where a single data tile can cover a larger portion of the lake, the MedAD is likely a better representation of model error due to minimal spatio-temporal differences. For instance, in the current study we found that the MedAD in LW was 24%, while Mishra *et al* (2023a) found that Lake Apopka, a small lake in Florida with an area of approximately 124 km², had a smaller MedAD of 15%. The MedAD values from Lake Apopka and LW indicate an expected model error range of 15% to 24%. Additional differences can be attributed to other factors (view angle, bio-optical variations, spatio-temporal changes, etc), as illustrated in figures S2–S6. However, further validation in smaller lakes is needed for a more robust assessment of the model error.

Excluding the model error itself, most of the differences between MODIS-CyanNet and MERIS/OLCI were due to: (1) lower sensitivity of CyanNet at the lower end of the biomass ($<2 \times 10^4$ CI_{cyno}, equivalent to a *Microcystis* cell concentration of 20 000 cell ml⁻¹), (2) differences in temporal and spatial satellite coverage within the 10 day composites for each of the satellite missions, and, less significantly, (3) mismatch due to pixel flagging as valid/invalid data (clouds, sun glint, land adjacency, etc) (Mishra *et al* 2023a). Differences in spatial resolution (OLCI's 300 m vs MERIS/MODIS's 1.1 km) may also have contributed to the mismatch. In the current work, we observed that the apparent overestimations by CyanNet were due to the difference in spatio-temporal satellite coverage. For example, during the peak of the 2011 blooms, MERIS missed three bloom days, 11 August 2013 Aug, and 16 Aug, when the biomass was very high. The additional observations by MODIS led to better characterization of the bloom by gap-filling with MODIS-CyanNet. However, the comparison analysis would show overestimation by CyanNet in the 09 August–18 August 2010-day composite (figure S2). Similarly, in the 19 August–28 August composite in 2011, imagery from 19 Aug was the defining bloom day with high biomass, including possible surface scum, spanning much of WLE. This day was missed by MERIS but captured by MODIS (figure S3). These two composites underscore that many of the differences between MERIS and MODIS are due to differences in temporal observations, and not due to error or overestimation by CyanNet.

Before the launch of S3B in 2018, OLCI also missed some of the bloom days, leading to apparent overestimation by CyanNet. For example, in the 08 September–17 September 2017 composite, the last three days were the defining bloom days with very high biomass in WLE. Although OLCI captured the bloom on 15 September, it missed the bloom on 16 September and 17 September while the bloom was intensifying in both biomass and extent (figure S4).

On the other hand, significant underestimation by CyanNet can be explained by either missing days or due to cloud coverage in MODIS, as documented in Lake Okeechobee and Lake Apopka, FL (Mishra *et al* 2023a). We observed the same problem in the Great Lakes region. Significant underestimation by CyanNet in the 19 August–28 August 2009 composite was primarily due to the bloom on 24 August, which MERIS captured better than MODIS-CyanNet (figure S5). Similarly, in the 29 August–7 September 2021 composite, clouds obscured the bloom on 29 August along the Ohio shoreline in the MODIS image. Additionally, the bloom on 31 August was captured by OLCI but missed by MODIS. The bloom then intensified on 05 September, which was well captured by OLCI, but data from MODIS were not available for that day (figure S6).

4.2. Interannual changes in bloom intensity, extent, and occurrence across the Great Lakes Region

In Lake Erie (WLE + CLE), the three largest bloom years included 2011 (6999 km²), 2015 (5409 km²), and 2013 (3456 km²). In 2011 and 2015, the bloom expanded into the CLE. The bloom extents were consistent with previous studies (Stumpf *et al* 2016b, Zeng and Binding 2021). However, the 2011 bloom was reported to be the largest bloom of the time with a bloom extent of 6000 km², which appears to be underestimated due to the lack of complete coverage of the bloom by MERIS.

While a long-term trend from 2000 suggests a steady increase in blooms in WLE, the results were more nuanced with substantial interannual variability. While blooms in 2020–2024 were larger than from 2000–2007 (excepting 2003), overall, the most substantive blooms occurred from 2008–2019. Similarly, Sayers *et al* (2019) reported a change of 25.28 km² yr⁻¹ as a linear trend with SeaWiFS and MODIS-Aqua observations from 1998 to 2017. Their analysis ended with several of the most intense bloom years (e.g. 2011, 2015, 2017) and did not capture the more current conditions (2018–2024). While a linear trend is informative for overall trends, more attention is needed on the interannual

variations of the environmental forcing factors (Stumpf *et al* 2016b) that can lead to non-monotonic changes in bloom intensity and size.

Overall, the results reported in this study are consistent with previously published reports of cyanohAB bloom intensity and extent in the Great Lakes region (Wynne *et al* 2021b, 2022). For example, Wynne *et al* (2021b) analyzed cyanobacterial bloom phenology in SGB using MODIS data and compared it with WLE. The study found that blooms in SGB occurred consistently at the same time and place each year, whereas WLE exhibited greater variability in bloom timing, location, and severity. SGB also showed minimal interannual variability in the intensity of the bloom, with the CI being within a factor of 2.5 (CI unit) in all years observed. GB also experienced large variability, consistent with Wynne *et al* (2022), who found that blooms in GB showed more interannual variability relative to WLE and SGB. Finally, SNB and LW consistently showed the highest bloom frequencies and spatial coverage across all regions, indicating persistent eutrophic conditions. This is likely due to the constrained size of these water bodies, relative to nutrient loads from their primary rivers. However, both SGB and GB showed moderate signs of improvement in bloom intensity, offering some hope for improving water quality.

4.3. Impact of Ballville Dam removal on SNB

We found a noticeable change in the 2019 annual mean bloom extent in SNB following upstream dam removal (figures 5(c) and (i)). The annual mean bloom extent decreased from 148 km² (mean bloom extent from the 3 years preceding the dam removal) to 98 km² (−29%, post-three-year mean) and 101 km² (−27%, post-six-year mean). Similarly, the post-three-year and post-five-year mean bloom intensities decreased 62% and 58%, highlighting the extent and intensity of cyanohAB blooms in SNB considerably decreased after 2019. This could be due to the impacts of flushing after the removal of the dam on the Sandusky River. Briefly, the Ballville Dam was built in 1911 on the Sandusky River to provide hydroelectric power to Fremont, Ohio (USFWS 2016). The dam was removed in October 2018 due to several pressing reasons, including acting as a physical barrier by restricting the upstream migration of various fish species such as the economically and ecologically important walleye (*Sander vitreus*; (Schulz *et al* 2025)). Following the removal of the dam, a rapid ecological shift occurred in SNB: the toxic cyanobacterium *Planktothrix* largely disappeared (Wagner *et al* 2024) and was replaced by *Cyanobium* spp. In combination with other potentially toxic genera, such as *Aphanizomenon*, *Dolichospermum*, and *Cylindrospermopsis*, indicating a dynamic and evolving biological landscape that necessitates continued monitoring for new toxins of concern.

4.4. Observed change in GB and LW

In GB, although the maximum bloom extent showed a decreasing trend from 2001 to 2009 (Sen's slope = −2.3 km² yr^{−1}), the trend was not consistent or meaningful (Kendall's tau = −0.09; figure 5(k), table S4). The bloom extent was the smallest in 2009, possibly due to the low detrital matter concentration in the bay (Wynne *et al* 2022). The lower absorption of Photosynthetically Active Radiation by detrital matter leaves the light available for photosynthetic activity by diatoms and green algae, allowing them to compete with cyanobacteria more effectively.

5. Summary and conclusions

We developed a multi-sensor time series for cyanobacterial blooms in the Great Lakes by incorporating a deep learning framework, CyanNet, to MODIS in combination with MERIS and OLCI, thereby filling gaps in the MERIS and OLCI time series. We demonstrated that CyanNet, an existing model, produces results consistent with MERIS and OLCI across the Great Lakes without retraining. Differences between CyanNet and MERIS/OLCI were not always due to model error, but were also attributed to the differences in satellite observation timing and coverage of the different sensors. Although training with new data may improve the model performance, the successful evaluation of CyanNet underscores the geographic transferability of the model, highlighting that model retraining is not necessary. We then used CyanNet to fill the CI_{cyno} observation gap between the Envisat (MERIS) and S3 (OLCI) missions. The result is 25 years (2000–2024) of continuous CyanohAB time series in the U.S. Great Lakes region.

Our cyanohAB timeseries analysis showed a tendency toward increased cyanohAB intensity and extent in the WLE basin since 2000. Conversely, we observed slight but decreasing trends in intensity in SNB and SGB. In SNB, these changes were associated with upstream dam removal in the late 2010s. Additional examination of this data set will improve our understanding of bloom response to changes in nutrient loads in these water bodies, including Lake Erie. The Copernicus OLCI mission is expected to

continue well into the future, with a third satellite, S3C, scheduled to launch in late 2026, followed by S3D two years later. This ongoing satellite data will enable us to compare current conditions with those from the early 2000s, extending our insights far into the future. Such long-term datasets are crucial for identifying subtle shifts and emerging patterns in bloom phenology that may otherwise go unnoticed.

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Data availability statement

The data that support the findings of this study are available as a NOAA National Centers for Environmental Information (NCEI) dataset (Mishra *et al* 2026) at <https://doi.org/10.25921/wzk1-r208>.

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Conflict of interest

The authors do not declare any conflict of interest.

Ethics statement

This research did not involve human or animal participants.

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